

Original papers

Apple crop-load estimation with over-the-row machine vision system

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ABSTRACT

Accurate crop-load estimation is important for efficient management of pre- and post-harvest operations. This information is crucial for the planning of labor and equipment requirement for harvesting and transporting fruit from the orchard to packing house. Current machine vision-based techniques for crop-load estimation have achieved only limited success mostly due to: (i) occlusion of apples by branches, leaves and/or other apples, and (ii) variable outdoor lighting conditions. In order to minimize the effect of these factors, a new sensor system was developed with an over-the-row platform integrated with a tunnel structure which acquired images from opposite sides of apple trees. The tunnel structure minimized illumination of apples with direct sunlight and reduced the variability in lighting condition. Images captured in a tall spindle orchard were processed for identifying apples, which achieved an identification accuracy of 79.8%. The location of apples in three-dimensional (3D) space was used to eliminate duplicate counting of apples that were visible to cameras from both sides of the tree canopy. The error on identifying duplicate apples was found to be 21.1%. Overall, the method achieved an accuracy of 82% on estimating crop-load on trees with dual side imaging compared to 58% with single side imaging. Over-the-row machine vision system showed promise for accurate and reliable apple crop-load estimation in the apple orchards.

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1. Introduction

Crop-load estimation is essential for efficient and effective management of orchards at various stages during the production cycle. In early summer, crop-load estimation (including counting and location of apples) helps to optimize green fruit thinning (Volz, 1988), which is critically important to improve fruit quality and yield. With this information, a producer can develop more sound crop load management and harvest strategies and can maximize their profits (Mizushima and Lu, 2011). It also helps producers to insure their crops (Cohen et al., 2011) so that they can be compensated in the event of financial loss due to weather related crop loss. Further, crop-load estimation prior to harvest is essential for efficient management of the labor force, harvest equipment, and vehicles for transportation of fruit from field to the packing plant. Packing house can also be benefited by optimizing postharvest handling process and storage with early crop-load information (Cohen et al., 2011).

Accurate crop-load estimation has always been a challenge for tree fruit producers (Winter, 1986). Several forecasting models

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have been developed based on ecological and/or cultivar related parameters for yield prediction (Stajanko et al., 2004). One such example is the 'Prognosfruit' crop forecasting model, which estimates the number of apples per tree based on yield capacity of tree, fruit density and average fruit mass (Winter, 1986). This method is time consuming, as different parameters have to be measured or estimated for individual orchards (Stajanko et al., 2004). Researchers have also investigated the development of yield forecasting models based on vegetation indices estimated with hyperspectral or multispectral images captured by areal platforms (Best et al., 2008; Ye et al., 2007). Ye et al. (2007) predict citrus yield from hyperspectral images using vegetation indices (e.g. NDVI, SR and PRI) and Partial Least Square (PLS) regression model. However the accuracy of these methods has been limited due to climate variabilities, cultivars and geographic locations (Aravena Zamora et al., 2010). Crop-load estimation based on direct counting of fruits in selected trees is another approach. However, manual countings are generally based on random sampling within trees, which require high labor input. Manual sampling may also have low statistical efficiency and could be biased (Aravena Zamora et al., 2010). In order to avoid such a bias on estimation, Wulfsohn et al. (2012) classified the tree rows based on vegetative indices to develop multi-level systematic (manual) sampling techniques to estimate fruit counts. But, manual sampling is time

consuming and has chances of inaccuracy due to manual error (Linker et al., 2012).

To address these issues, machine vision systems involving different types of sensors and image processing techniques have been investigated. Aggelopoulou et al. (2011) used a color camera to identify flowers in trees, which was then correlated to crop-load. In such a system, fruit set, thinning and fruit drop later in the season (Winter, 1986) will complicate the estimation and affect the accuracy of estimation. To estimate crop-load based on identification of individual fruit, various studies have been carried out using different types of sensors like color camera (Linker et al., 2012; Tabb et al., 2006), thermal camera (Stajanko et al., 2004), and multispectral and hyperspectral cameras (Kim and Reid, 2004; Best et al., 2008; Safren et al., 2007). Further, various types of image processing techniques have been investigated to identify fruit, which can roughly be divided into shape-based and spectral-based analysis (Bulanon and Katoka, 2010). Bulanon et al. (2002) used spectral analysis to identify apples and reported 88% accuracy. Linker et al. (2012), Stajanko et al. (2009), and Ji et al. (2012) used shape, color and texture based analysis and reported 85%, 89%, and 89% accuracy respectively. Fruit detection accuracy in those studies was limited by clustering and occlusions of fruits (Stajanko et al., 2009), and variable lighting conditions in orchards (Ji et al., 2012; Linker et al., 2012).

Most of these studies, however, evaluated the accuracy based on the number of apples visible in an image taken from one side of the tree canopy, which may be substantially different from the total number of apples in a tree. To address this issue, Wang et al. (2013) used a stereovision camera at night for multiple side imaging and compared the results with the actual number of apples counted by humans. This study achieved 60% accuracy in crop-load estimation when substantial number of apples were in clusters. Similarly, Vision Robotics Corp (San Diego, CA) reported an accuracy of 40–60% without correction for non-visible apples in images (Koselka, 2010). To improve the accuracy, the study correlated the manual apple count with the vision-based estimated count. Zhou et al. (2012) also identified red and green apple using color information and correlated that to the ground truth with a correlation coefficient of 0.58–0.71. However, this process will be labor intensive and specific to particular type of orchard. Correlation coefficient may also change from year to year. Therefore, it is valuable to develop a machine vision system that can improve crop-load estimation accuracy to an acceptable level without correlating estimated count and ground truth collected by human.

Fruit identification in an orchard environment through image analysis is affected by various factors including occlusion of fruit, variable outdoor lighting conditions, complex plant structures, and irregularity of fruit shape and size (Cohen et al., 2011; Karkee and Zhang, 2012). As discussed before, the two most critical factors limiting the accuracy of crop-load estimation in apple orchards are; (i) a significant number of apples are invisible to cameras looking from only one side of apple tree canopies; and (ii) apple identification accuracy is compromised by variable lighting conditions. It was observed in the field that only 60% of apples were visible from one side of the canopy even in a modern tall spindle apple orchard because of occlusion of apples by leaves, stems and other apples. In addition, apples can be partially occluded or can be in clusters of two or more fruit, both situations are challenging for image processing techniques. Varying lighting conditions, on the other hand, causes non-uniform distribution of light intensity in apples based on their exposure to sunlight at the time of imaging, which may cause improper and incomplete image segmentation. Thus, it would be challenging to develop a robust algorithm to identify apples in variable lighting conditions. In this study, our focus is to minimize the effect of variable lighting conditions and occlusion of apples to improve apple crop-load

estimation in orchard environment. The major goal of this study was to investigate the potential of improving crop-load estimation with dual-side imaging compared to single-side imaging. The specific objectives of the study were:

1. To identify and count total number of apples in the images captured from opposite sides of tree canopies using an Over-the-Row (OTR) sensor platform; and
2. To use 3D location information to estimate number of apples visible from both sides and to avoid duplicate counting.

2. Materials and methods

An over-the-row (OTR) sensing platform with dimensions 2.13 m (length) $\times$ 2.74 m (width) $\times$ 3.67 m (height) was developed and mounted on the three-point hitch of a tractor so that it could be lifted up from one trees to another and dropped down to the ground while taking images. A color camera (Fig. 1a) (Prosilica GigE 1290c, Allied Vision Technologies, Stadtroda, Germany) with a field of view (FOV) of 43.6° (horizontal) by 33.4° (vertical) and image resolution of 1280 $\times$ 960 was used to acquire RGB (color) images. Also, a time-of-flight-of-light-based 3D camera (Fig. 1a) (PMD CamCube 3.0, PMD Technologies, Siegen, Germany) with image resolution of 200 $\times$ 200 and FOV of 40° (horizontal) and 40° (vertical) was used to obtain 3D information of objects. Two cameras were integrated together and mounted in the middle (lengthwise) of the platform with sliding mechanism so that camera height could be adjusted to capture the whole tree. To block direct sunlight while imaging, a tunnel (Fig. 1b) was created by covering the platform with tarpaulin (opaque curtain) on four sides and the top of the platform. A number of LED lights (Fig. 1c) (Trilliant® 36 Light Emitting Diode Grote, Madison, Indiana) were installed in the platform to create a controlled, uniform lighting environment inside the tunnel. The lighting system also added capability for nighttime data collection.

Four major steps were involved in identifying and counting apples while avoiding duplicate counting: (i) data collection, (ii) apple identification, (iii) co-registration of two dimensional (2D) color images and 3D images, and (iv) identification of duplicate apples and apple counting (Fig. 2). These steps are described in details in the following subsections.

2.1. Data collection

Canopy images were acquired from two sides of a row in commercial apple orchards using an over-the-row sensing platform. Apple trees used in this work were of 'Jazz' variety (Fig. 3) trained in the tall spindle architecture (row spacing 2.74 and inter-plant spacing 1.17 m) (Yakima Valley Orchards, Prosser, WA). Images were acquired between four and one week before harvest season in 2013. During the earlier weeks of data collection, the apples were immature and green in color. During the last week of imaging, apples matured and acquired their full size and red color with good hue and intensity. The color and 3D cameras used in this work had relatively small field-of-view, which were not enough to capture the whole tree canopy at once. To cover entire canopies, images were acquired from five different camera heights (Fig. 4). The over-the-row platform with sensing system was driven through apple tree rows to capture images of 20 trees. A total of 212 canopy images were captured with 3D camera and another 212 images were captured by the color camera (424 images in total). Out of 212 color images, 104 images were captured when apples were still immature and 108 images were captured from trees with mature fruit. Two people counted the total number of apples in each tree manually for the ground truth dataset. Actual number of apples that were visible in the images captured from both sides

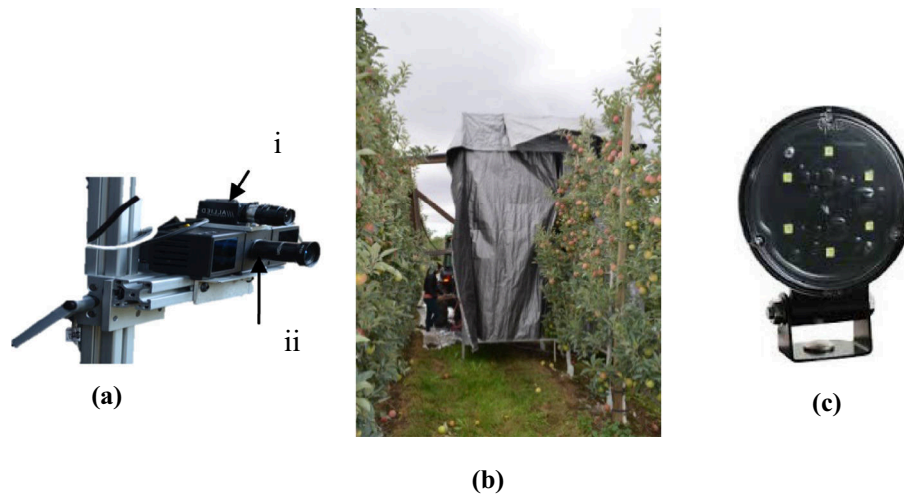


Fig. 1. Sensor system used for data collection: (a) Prosilica GigE Color Camera (i) and PMD CamCube 3D camera (ii), (b) Over-the-Row (OTR) platform with tunnel structure, and (c) Trilliant® 36 LED light. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

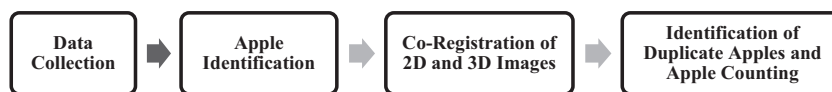


Fig. 2. Functional block diagram of the apple identification and counting system.



Fig. 3. (a) A commercial Jazz apple orchard in tall spindle architecture in Prosser, WA (Yakima Valley Orchards, Prosser, WA) and (b) over-the-row platform taking images.

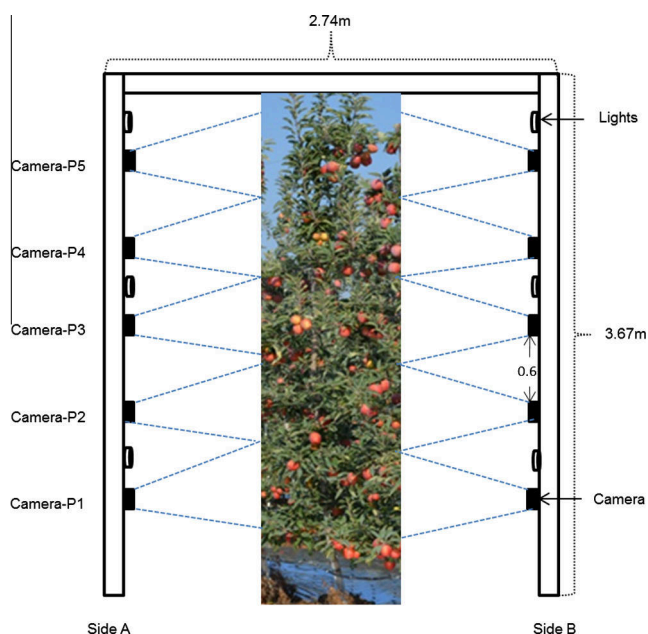


Fig. 4. Cross-sectional view of platform and sensor system.

of the canopy (called ‘duplicate apples’ in this paper) were counted manually in the images.

2.2. Apple identification

The apple identification method used in this work was based on color and shape features of apples as explained in [Silwal et al. \(2014\)](#). The method is briefly described here as follows ([Figs. 5 and 6](#)). First, histogram equalization was applied to color images in Hue, Saturation and Intensity (HSI) color space, which intensified color differences between apples (foreground), and leaves and branches (background). To remove noises that were intensified during histogram equalization, a pixel wise adaptive Wiener filter ([Gonzalez et al., 2010](#)) of neighborhood size 5×5 was applied. This filter minimizes the mean square error in the overall process of inverse (high pass) filtering and noise smoothing so that the output present the optimum linear estimation of original image. Images were then converted back to red, green and blue (RGB) color space, and a grayscale image was created by subtracting green and blue channels from red channel. These gray scale images were converted to binary images using Otsu’s threshold method ([Gonzalez et al., 2010](#)), which determines an optimal threshold value to separates image pixels into foreground and background classes with minimum intra-class variance. To remove small-unwanted objects

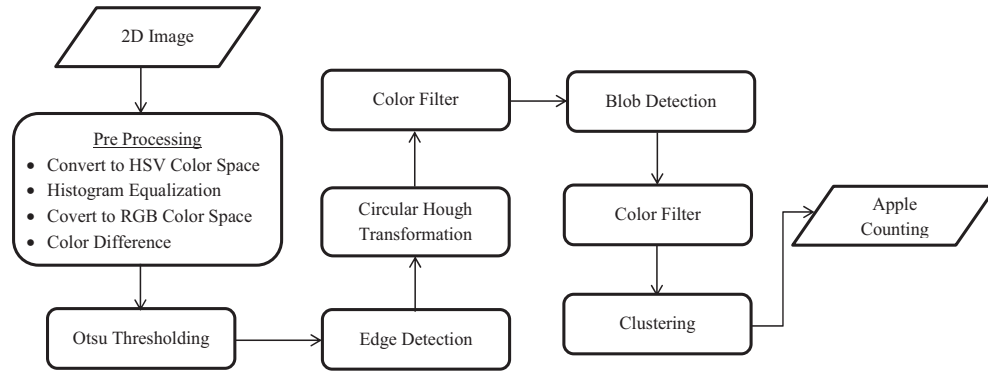


Fig. 5. The Functional Block Diagram of Apple Identification Process.

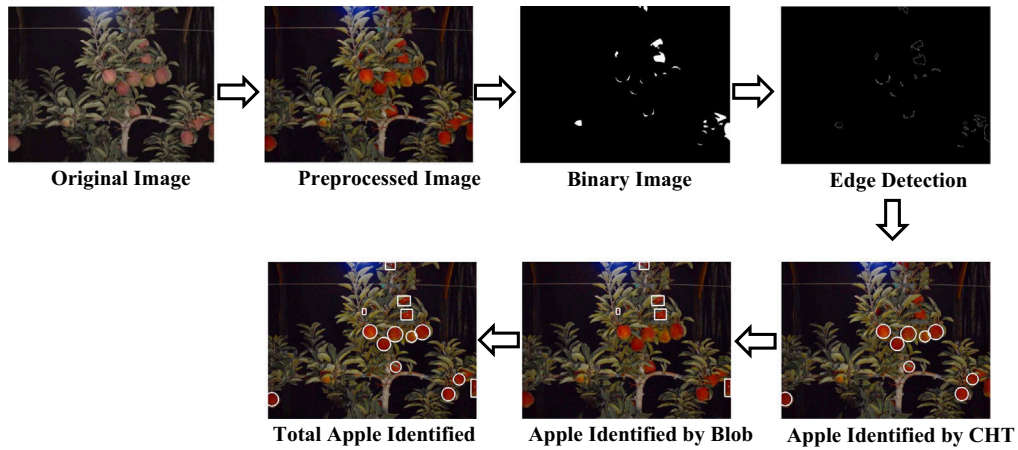


Fig. 6. Apple identification process depicted with a sample image.

segmented as foreground, regions with area less than 100 pixel and compactness less than 190 were filtered out. Edge detection was then carried out using Canny edge detector for its ability in detecting weak and strong edges separately using two different thresholds (Gonzalez et al., 2010).

To identify circular objects that are likely to be apples, Circular Hough Transformation (CHT) was applied in an iterative fashion within a specified range of radius. As the color camera with $43.6^\circ \times 33.4^\circ$ FOV was located approximately 0.9–1.5 m away from the center of the tree canopies, image size of apples with 6–8 cm physical diameter varied approximately from 35 to 65 pixels. Therefore, radius of the circle used in iterative CHT varied from 30 pixel to 65 pixel. Then, color information was used to filter out objects identified by CHT that did not have desired level of redness so that false positive apple identification could be minimized.

Finally, blob analysis (Moeslund, 2012) was carried out using color information to identify partially visible apples that were not identified by CHT. Blob analysis also used area and compactness to analyze the consistency of fragments of apple. Blobs with area above 90 pixels and compactness above 90 were considered a part of the apple(s). To avoid detections of several fragments of an apple as multiple apples, a clustering algorithm based on Euclidian distance between the blobs was used. If blobs were within the circumference of a circle with a radius of 65 pixel (largest apple identified by CHT), they were considered parts of the same apple. This apple identification method was used to identify apples in all images captured from multiple locations in both sides of tree canopies and identification accuracy was assessed using apples counted manually in those images. Image

processing algorithms were implemented in MATLAB software (R2012a, Math Works Inc., Natick, MA).

2.3. Co-registration of 2D and 3D images

A color camera and a 3D camera were mounted on top of each other to acquire color and 3D images together with a similar field of view. To co-register images captured by these two cameras, a camera calibration toolbox developed by MATLAB (Bouguet, 2013) and a checkerboard ($0.045 \text{ m} \times 0.05 \text{ m}$) were used. The checkerboard was placed at a distance of 1.5 m in front of the imaging system and 25 images of checkerboard were captured by both 3D and color camera at varying orientations (examples are shown in Fig. 7). The intensity images from the 3D camera and images from color camera were used to extract grid corners in checkerboard using calibration toolbox and then intrinsic camera parameters were determined (Table 1). These intrinsic parameters included focal length (F), principle point (C) and distortion coefficient (k) (skew coefficient, radial and tangential distortions). Using these intrinsic parameters, a stereo calibration was performed to determine extrinsic camera parameters, which were rotation (R) and translation (T) vectors of the 3D camera with respect to the color camera.

In addition, the intrinsic parameters of camera were used to transform the coordinates from camera frame to the image plane using pinhole model projections, also called normalized imaged projection (Bouguet, 2013). In Fig. 8, $P [X; Y; Z]$ is a point in 3D space, and $P_N [X_N; Y_N]$ is the normalized pinhole projection of point

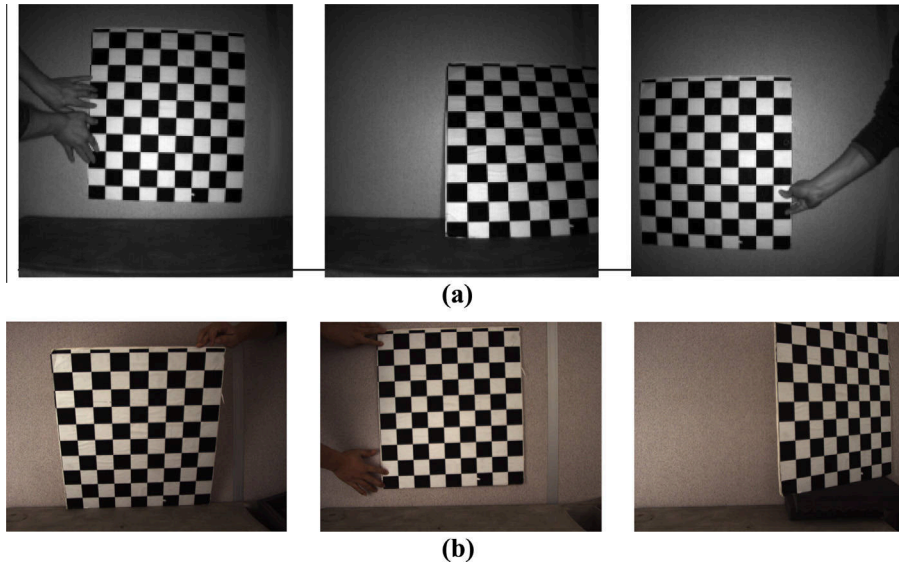


Fig. 7. 3D camera intensity image (a), and 2D camera color images (b) of a checkerboard captured at different orientation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
The intrinsic camera parameter of PMD and 2D camera.

	Focal length		Principle point		Distortion coefficient				
	F_x	F_y	C_x	C_y	K_1	K_2	K_3	K_4	K_5
PMD camera	287.3	287.8	103.3	99.0	−0.44	0.22	−0.00042	−0.00273	0
2D camera	1641.3	1642.1	620.7	478.7	−0.17	0.16	0.00035	−0.00088	0

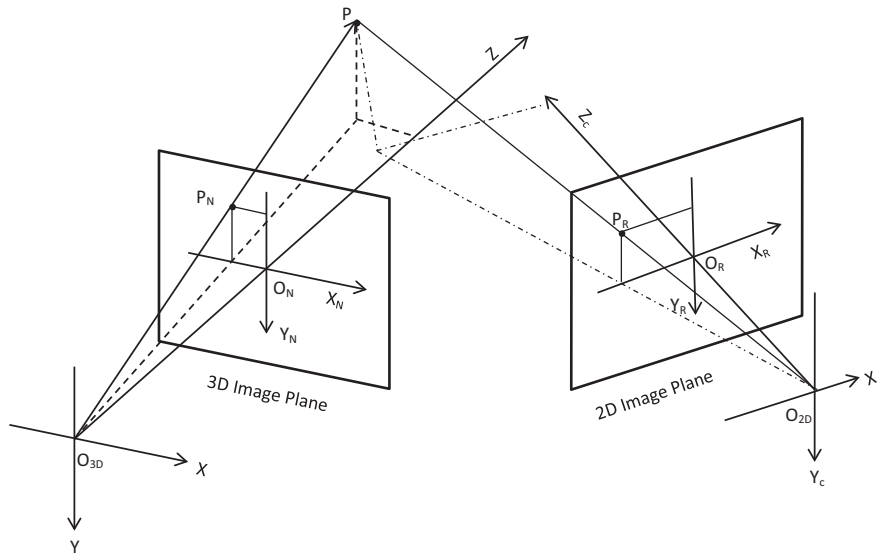


Fig. 8. Pinhole camera model for 2D and 3D camera.

P (after distortion correction) on the image plane, which is given by Eq. (1).

$$\begin{bmatrix} X_N \\ Y_N \\ 1 \end{bmatrix} = K \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad (1)$$

$$\text{where } K = \begin{bmatrix} F_x & 0 & C_x \\ 0 & F_y & C_y \\ 0 & 0 & 1 \end{bmatrix}.$$

F_x , F_y , C_x and C_y are focal distances and principle points respectively of the PMD Camera.

In this study, to co-register the image from 3D and 2D cameras, inverse mapping of point P_N from 3D image plane to point P in 3D space was done to determine the coordinates of objects in 3D space (Bouguet, 2013; Reulke, 2006). Extrinsic camera parameters were then used for rigid motion transformation of point P [X ; Y ; Z] from 3D camera reference frame to P_c [X_c ; Y_c ; Z_c] in color camera reference frame using Eq. (2) (Szeliski, 2010). Then, the point P_c in 3D

space was projected to color image plane $P_R [X_R; Y_R; Z_R]$ using normalized pinhole projection method (Eq. (1)). This process was repeated to get coordinates of all the points in 3D image with respect to 2D camera reference frame.

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = R \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + T \quad (2)$$

where $R = \begin{bmatrix} 0.999 & 0.016 & -0.031 \\ -0.016 & 0.999 & -0.023 \\ 0.029 & 0.024 & 0.999 \end{bmatrix}$ rotation matrix and, $T = \begin{bmatrix} 3.176 \\ 60.50 \\ 18.11 \end{bmatrix}$ is the translation vector derived through camera calibration.

2.4. Identification of duplicate apples and overall apple counting

Images of apple trees were acquired from opposite sides of the tree canopy from multiple locations. Some apples were visible in the images captured from multiple locations on the same side. More importantly, a substantial number of apples were visible in both images captured from opposite sides of the canopy, which will lead to an overestimation of crop-load because of duplicate counting. To avoid this issue, identification of duplicate apples is crucial in multi-side imaging. The flowchart of the steps involved in identification of duplicate apples is shown in Fig. 9.

Apples identified in multiple images were mapped together in a common coordinate system to identify the duplicate apples. Apples identified in color images were masked and overlaid on respective 3D coordinate images. Then 3D coordinates of each pixel within the apple regions with respect to the 2D camera coordinate system (called local coordinate system) were determined. Because images were acquired from different camera locations, 3D transformation of coordinates was needed to map apples on to a common coordinate system (called a global co-ordinate system). The planar, rectangular geometry of the sensor platform, and relative camera positions and orientations were used to determine the parameters for 3D coordinate transformation. To transform coordinates from one side of the canopy with respect to the other, it was assumed that cameras were located at the same height and on the same plane in two sides of the platform (to be called Side A and Side B), therefore, requiring no translation in y-coordinates. The translation in Z axis (direction of depth) was done by subtracting z-coordinates of objects from the distance between two cameras. For example, in Fig. 10, if Z coordinate of a point a with respect to Side A (Z_a) is 20 cm and the distance (D) between two camera locations A and B is 80 cm, the location of the point a' in z-axis with respect to Side B will be 60 cm ($D - Z_a$).

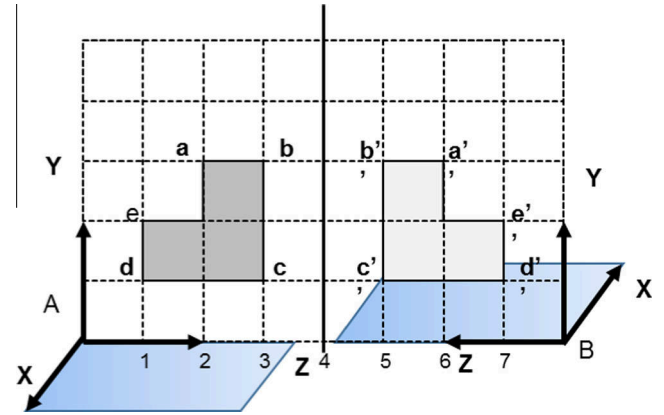


Fig. 10. Transformation of object coordinates abcde with reference to Side A to the new coordinates a'b'c'd'e' with respect to Side B after translation in Z-axis and reflection in X-axis.

Additionally, as two cameras were facing each other, the direction of x-axes for these cameras are opposite to each other as represented in Fig. 10. Therefore, reflection about the x-axis was carried out to obtain x-coordinates of objects with respect to camera on the opposite side. This translation and reflection of objects to transform the coordinate system can be represented by Eq. (3). All the points in 3D point cloud from two sides were transformed to the same coordinate system using this transformation.

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} Xb \\ Yb \\ D - Zb \end{bmatrix} \quad (3)$$

However, due to uneven terrain, and ground conditions, and the unstable structure of the sensor platform, relative camera positions and orientations did vary slightly when the platform moved from one place to the other in the orchard. Therefore, starting with the ideal transformation parameters discussed in previous paragraph, a trial and error method was used to derive more accurate transformation parameters. Five apples visible from both sides of tree canopy were used in each apple tree to carry out trial-and-error method. The 3D transformation parameters obtained from this method varied slightly between datasets collected from different locations primarily because of the instability of the platform. Transformation parameters used were 0.05 m of translation in X-axis and Y-axis, and 2.4 m or 2.45 m in Z-axis depending on the location.

After 3D transformation to a common coordinate system, the distance between each apple captured from opposite sides at the same height were determined by using coordinate of apple centers. Since there were inaccuracies in sensor data, errors in apple identification, and error in 3D registration methods, 3D locations

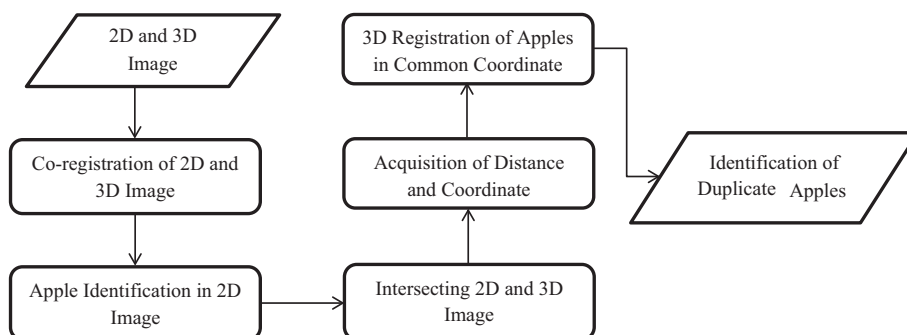


Fig. 9. Flowchart of duplicate apple identification process.

of two instances of the same apple (imaged from different locations) were displaced slightly. In addition, some closely positioned apples were overlapped in the registered 3D space. Thus, buffer distance of three-fourth of average diameter of apple (0.075 m in this study) was used to minimize error in identifying duplicate apples. Apples that were located at a distance less than the buffer distance were considered as the duplicate apples. Also, false positives (apples that were not duplicated but were identified as duplicate) and false negatives (actually duplicated apples but not identified as duplicate) were determined. A confusion matrix was also created to compare the apple identification results with the ground truth in terms of producer accuracy and consumer accuracy. Producer accuracy provided the probability that all duplicate apples are identified as duplicate apples, and consumer accuracy provided the probability that the identified duplicate apples are actually duplicate apples.

To estimate total apple count in each tree, numbers of duplicate apples were subtracted from the total apple counts identified in images captured from various locations. Accuracy of the fruit counting method was then assessed by comparing the final count with the number of fruit counted manually in the field. Two people counted the number of apples in each tree during field data collection to minimize potential human error. Average of these two counts was used as the reference data for comparison. A comparison was also made between the accuracy of apple counting with dual side imaging and that with single side imaging. These comparisons were made using Mean Absolute Percentage Error (MAPE) calculated using Eq. (4).

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \frac{|n_i - n_c|}{n_i} \quad (4)$$

where

- n_i = apples identified by algorithm in each tree,
- n_c = manually counted apples in each tree, and
- N = total number of tree samples.

3. Results and discussion

3.1. Apple identification

The integrated method with color segmentation, CHT and blob analysis was able to identify apples with fully developed red color

Table 2
Accuracy of apple identification method.

S.N.	Total Apples Side A		Total Apples Side B		False positive	False negative	Absolute error (%)	
	Actual	Identified	Actual	Identified			Side A	Side B
1	66	35	75	54	5	57	47.0	28.0
2	49	24	54	31	2	49	51.0	42.6
3	89	45	94	48	6	96	49.4	48.9
4	73	39	81	47	10	78	46.6	42.0
5	97	61	104	69	11	82	37.1	33.7
6	56	52	67	46	11	36	7.1	31.3
7	64	47	56	46	5	32	26.6	17.9
8	67	41	71	53	13	57	38.8	25.4
9	41	45	39	41	34	28	9.8	5.1
10	65	56	59	50	23	41	13.8	15.3
11	80	62	79	73	17	41	22.5	7.6
12	74	62	92	78	12	38	16.2	15.2
13	39	34	41	31	7	22	12.8	24.4
14	75	68	86	61	12	44	9.3	29.1
15	53	45	64	55	7	24	15.1	14.1
16	62	56	67	60	12	25	9.7	10.4
17	71	67	60	56	20	28	5.6	6.7
18	63	60	54	56	16	18	4.8	3.7
19	63	62	63	59	20	25	1.6	6.3
20	77	73	75	68	11	22	5.2	9.3
MAPE							21.17	

as well as greenish immature apples (Fig. 6). The mean absolute percentage error in identifying all apples was found to be 21.2% (Table 2). Both false negatives (missing actual apples) and false positives (identifying noises such as stems, branches and leaves) were present in the result. There was 13% of false positive identification, which primarily included branches and leaves with similar color intensities to that of apples (Table 3). Because the apple identification algorithm was optimized for red apple identification, substantial numbers of greener; immature apples were missed leading to 33.2% false negatives. When images of trees with greenish apples captured three to four weeks before harvest were neglected, the mean absolute percentage error reduced to 12%, with 13.9% false positive identification and 25% false negatives. Further improvement in the apple identification method will be necessary to increase the accuracy of apple identification, particularly for green apples. However, the major goal of this work was to investigate the potential of improving crop-load estimation with dual-side imaging compared to single-side imaging and therefore this level of accuracy was considered sufficient for proving the concept.

3.2. Identification of duplicate apples

Integration of color and 3D camera images provided locations of apples in 3D space (Fig. 11). After 3D transformation of coordinates from individual camera locations to the common global coordinate system, apples were plotted as spheres in a 3D plot (Fig. 12). The mean absolute percentage error (MAPE) in identifying duplicate apples was estimated to be 21% (Table 4).

A confusion matrix based on all the apples used in evaluating duplicate apple counting algorithm is presented in Table 5. Out of the total duplicate apples identified by the algorithm, 93.8%

Table 3
False positive and false negative on apple identification.

		Predicted		False negative
		Apples	Not apples	
Actual	Apples	1697	843	33.2%
	Not apples	254		
	False positive	13.0%		

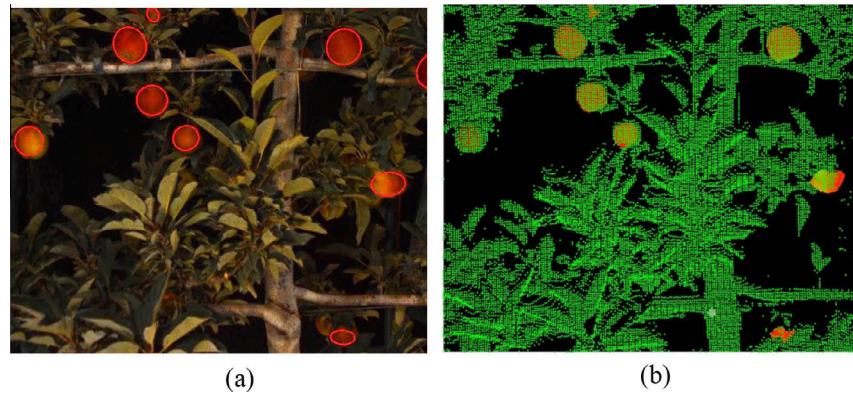


Fig. 11. (a) Apples identified in a color image and (b) corresponding apples in the 3D image. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

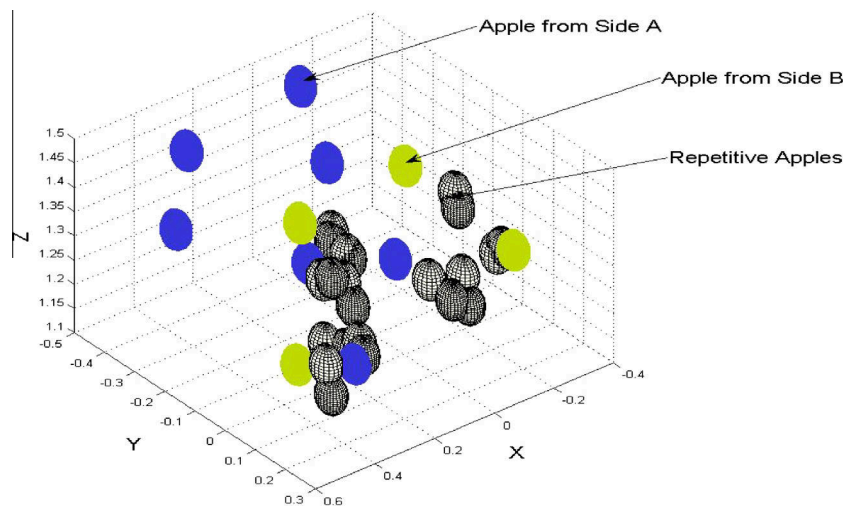


Fig. 12. Mapping of apples identified from two sides of an apple tree canopy in a global 3D coordinate space. In general, overlapping apples represent the same apple imaged from the two sides of the same tree canopy.

Table 4
Accuracy of identifying duplicate apples.

Tree No.	Duplicate apples		False Positive	False Negative	Absolute error (%)
	Actual	Identified			
1	13	14	1	0	7.7
2	5	3	1	3	40.0
3	10	4	0	6	60.0
4	9	7	0	2	22.2
5	24	22	0	2	8.3
6	16	12	2	6	25.0
7	23	17	0	6	26.1
8	6	5	0	1	16.7
9	12	12	2	2	0.0
10	14	13	1	2	7.1
11	12	12	1	1	0.0
12	22	22	2	2	0.0
13	15	10	1	6	33.3
14	32	21	0	11	34.4
15	15	8	1	8	46.7
16	44	31	0	13	29.5
17	42	47	6	1	11.9
18	14	11	1	4	21.4
19	33	25	0	8	24.2
20	48	44	2	6	8.3
MAPE					21.1

Table 5
A confusion matrix for duplicate apple identification method showing consumer's and producer's accuracies.

		Predicted		Producer's accuracy
		Duplicate	Non-duplicate	
Actual	Duplicate	319	90	78.00%
	Non-duplicate	21	1896	95.47%
Consumer's accuracy		93.8%	98.9%	

were actually repetitive apples (Consumer's Accuracy). Similarly, accuracy for classifying duplicate apple as duplicate was 78.0% (Producer's Accuracy). The consumer accuracy represented false positives, which were mainly caused by the displacement of apples in 3D positions. When apples located close to each other or in clusters were displaced slightly toward each other, they were identified as duplicate apples. Further improvement in 3D sensing and co-registration methods could potentially reduce false positive identification and overall accuracy of duplicate apple counting method.

Table 6
Apple counting error with dual side imaging of tree canopies.

S.N.	Apple count	Apple identified		Total (A + B)	Duplicate Apple (C)	Apple count	Absolute error (%)
	Actual	Side (A)	Side (B)			Identified (A + B – C)	
1	93	35	54	89	14	75	19.4
2	74	24	31	55	3	52	29.7
3	160	45	48	93	4	89	44.4
4	115	39	47	86	7	79	31.3
5	128	61	69	130	22	108	15.6
6	92	52	46	98	12	86	6.5
7	85	47	46	93	17	76	10.6
8	112	41	53	94	5	89	20.5
9	60	45	41	86	7	79	31.7
10	106	56	50	106	13	93	12.3
11	126	62	73	135	11	124	1.6
12	123	62	78	140	17	123	0.0
13	52	34	31	65	10	55	5.8
14	90	68	61	129	20	109	21.1
15	77	45	55	100	8	92	19.5
16	76	56	60	116	27	89	17.1
17	78	67	56	123	37	86	10.3
18	88	60	56	116	11	105	19.3
19	79	62	59	121	24	97	22.8
20	82	73	68	141	42	99	20.7
MAPE							18.0

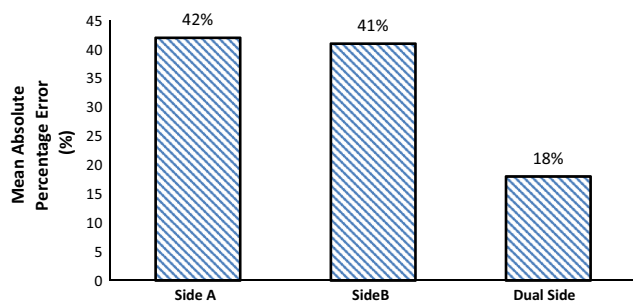


Fig. 13. Error on apple counting for single side imaging and dual side imaging.

3.3. Overall apple counting

The Mean Absolute Percentage Error (MAPE) on overall apple counting was estimated to be 18% for dual side imaging (Table 6). Similarly, (MAPE) on total apple counting when apple trees were imaged from only one side were 42% and 41% respectively from front (A) and back (B) sides (Fig. 13). Based on our preliminary observation, apple visibility was only around 60% from one side of the canopy in the tall spindle apple orchard used in this work. The lower MAPE in apple counting with dual side imaging compared to single side imaging was mainly due to substantial improvement in apple visibility.

Error in apple identification was the major factor for underestimation of apple counts in trees in this study. Error in identification of duplicate apples also affected the accuracy of apple count, but in a smaller proportion. In this study, there was relatively high false negative apple identification, which led to underestimation of apple count. False identification of non-duplicate apples as duplicate apples also caused some level of undercounting. However, the number of false positives on apple identification and false negative on identifying duplicate apples canceled their effects slightly. In future, systematic application of horticultural operations such as training, pruning, pollination, and thinning may be investigated to improve visibility, and reduced occlusion and clustering of fruits. Also, further research may be carried out to detect and reconstruct partially visible apples in modern fruiting wall orchards, which will provide opportunity for further improvement in overall apple

counting through improvement in apple detection as well as 2D–3D registration method. Further studies may also be needed to address engineering challenges of the platform for real time operation in sloped orchards as this study was carried out in a relatively flat orchard. It is also noted that this work focused on estimating crop-load when fruit have developed desired color and size. Crop-load estimation would have higher value if it can be conducted early in the season. However, accurate crop-load estimation would help pre- and post-harvest management in the orchard even if it is conducted just a few days prior to harvest.

4. Conclusions

In this work, apple trees were imaged from two opposite sides of tree canopy using an over-the-row sensor platform to minimize the effect of occlusion and variable lighting conditions in apple crop-load estimation. Controlled lighting environment for imaging both at day and night time was created using artificial lighting in the system. Apples were identified in the images based on shape and color with an accuracy of 78.9%. Then the apples identified from both sides of tree canopy were mapped together in a 3D space to avoid duplicate counting. Accuracy on identification of those duplicate apples was 79%. Accuracy on overall apple counting increased substantially from 58% with single-side imaging to 82% with dual-side imaging. The results can further be improved with the implementation of better image processing and classification techniques to identify both green (immature) and red apples as well as through improvement in 3D sensing and mapping techniques. For practical application of crop-load estimation technique in orchards, sizing of apples and real-time sensing and image processing capability will have to be added.

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